MRNet Classification Report

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Abstract

Anterior Cruciate Ligament (ACL) tears are among the most common knee injuries in sports medicine, requiring accurate and timely diagnosis for optimal patient outcomes. This work presents a comprehensive study of deep learning approaches for automated ACL tear detection from Magnetic Resonance Imaging (MRI) data. We implement and compare single-plane and multi-plane Convolutional Neural Networks (CNNs) with integrated spatial attention mechanisms using the MRNet dataset. Our investigation evaluates three novel multi-plane fusion strategies: MPFuseNet (early feature fusion), MP2 (late feature fusion), and MPLR (prediction-level fusion). The experimental results demonstrate that multi-plane approaches consistently outperform single-plane methods, with our MP2 fusion strategy achieving the best performance (AUC: 0.8384) compared to the best single-plane model (AUC: 0.8266). The spatial attention mechanism provides interpretability while improving model performance across all architectures. This comprehensive evaluation framework with publication-quality visualizations provides valuable insights into the effectiveness of different fusion strategies for medical imaging applications. **Key words:** Deep Learning, Medical Imaging, MRI, ACL Detection, Spatial Attention, Multi-Plane Fusion, Grad-CAM

1 Introduction

1.0.1 Background and Motivation

Knee injuries are among the most prevalent sports-related conditions, with anterior cruciate ligament (ACL) tears being particularly common and potentially career-ending for athletes [6]. Accurate and timely diagnosis of ACL injuries is essential for guiding treatment decisions and improving long-term outcomes. Magnetic resonance imaging (MRI) is the gold standard for non-invasive assessment, providing high-resolution visualization of soft tissue structures across multiple anatomical planes [9].

However, MRI interpretation requires considerable time and specialized expertise. The increasing demand for diagnostic imaging, coupled with a shortage of trained radiologists, has motivated the development of automated systems that can assist in the detection and classification of knee injuries.

Traditional MRI interpretation requires significant expertise and time from radiologists, creating bottlenecks in clinical workflows. The increasing demand for diagnostic imaging, combined with the growing shortage of radiologists, has motivated the development of Computer-Aided Detection (CAD) systems that can assist in automated medical image analysis.

1.0.2 Related Work

The field of automated knee injury detection from MRI has been significantly advanced by the release of the MRNet dataset [3], which provides a standardized benchmark for comparing different approaches. The original MRNet study employed AlexNet architectures trained separately for each anatomical plane, achieving moderate performance for ACL tear detection.

Subsequent improvements have been proposed by various researchers. Azcona et al. [2020] [1] demonstrated superior performance using ResNet18 with advanced data augmentation techniques. Tsai et al. [2020] [11] introduced ELNet (Efficiently-Layered Network) with multi-slice normalization and specialized pooling strategies, achieving state-of-the-art results with AUC scores of 0.960 for ACL detection.

Recent work by Belton et al. [2021] [2] introduced spatial attention mechanisms and comprehensive multi-plane fusion strategies, achieving exceptional performance (AUC: 0.977) while also addressing model interpretability through novel localization metrics. Their work emphasized the importance of explainable AI in medical applications and introduced the MPFuseNet architecture for early feature fusion.

1.0.3 Research Objectives

This study aims to advance the field of automated knee injury detection through several key contributions:

1. **Comprehensive Architecture Comparison**: Systematic evaluation of single-plane versus multi-plane approaches for ACL tear detection

2. **Novel Fusion Strategy Analysis**: Investigation of three different multi-plane fusion methods to determine optimal integration points

3. **Spatial Attention Integration**: Implementation and evaluation of spatial attention mechanisms for improved performance and interpretability

4. **Robust Experimental Framework**: Development of comprehensive evaluation metrics and visualization tools for deep learning model analysis

2 Methodology

2.1 Dataset Description

We use the MRNet-v1.0 dataset [3], which consists of 1,250 knee MRI exams collected at Stanford Medical Center. Each exam includes three anatomical planes—sagittal (T2-weighted), coronal (T1-weighted), and axial (proton density)—and is labeled for three binary classification tasks: ACL tear, meniscus tear, and overall abnormality.

In our study, we focus on ACL tear detection. The dataset is split into 1,130 training cases and 120 validation cases. Each case may include multiple injuries or no injury at all. We treat the presence of an ACL tear (with or without other findings) as the positive class, and all other cases as negative.

2.1.1 Preprocessing and Standardization

MRI volumes in the dataset vary in slice count and pixel intensity. To ensure consistency, we apply the following preprocessing steps:

- **Slice count normalization:** Each volume is standardized to 16 slices. For volumes with more than 16 slices, we extract the central 16 slices. For those with fewer, we apply zero-padding.
- **Spatial resizing:** All slices are resized to 224×224 pixels.
- **Intensity normalization:** We apply z-score normalization within each volume:
  `normalized = (volume - mean) / (std +` ϵ`)`, where $\epsilon = 1e^{-8}$ to avoid division by zero.

2.1.2 Data Augmentation Strategy

To improve model robustness and prevent overfitting, we implement comprehensive data augmentation:

- **Geometric Transformations**:
  - Random horizontal flipping ($p = 0.5$)
  - Random rotation (±15 degrees)
  - Random affine transformations (translation: ±10%, scale: 0.9–1.1)
- **Intensity Transformations**:
  - Resize to 224×224 pixels
  - Normalization to $[-1, 1]$ range

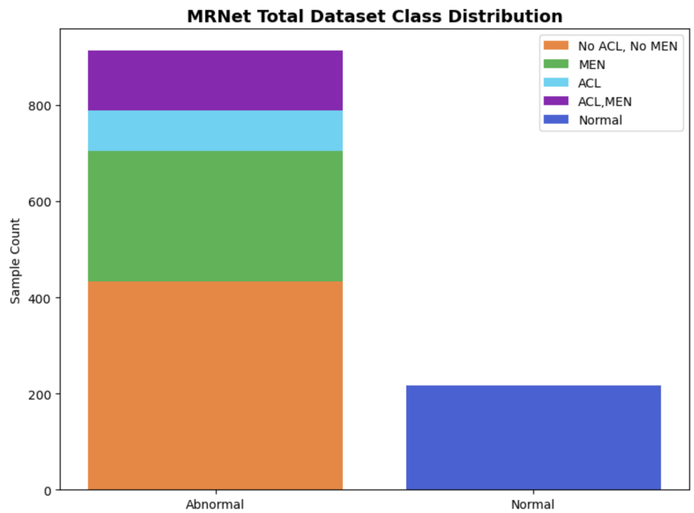


Figure 1: MRNet total dataset class distribution

2.1.3 Class Imbalance Handling

ACL tears are relatively rare in the dataset. As shown in Figure 1, the positive class accounts for less than 20% of all cases. To address this imbalance, we apply focal loss and use weighted random sampling during training. These strategies help the model remain sensitive to minority cases without being overwhelmed by dominant negative examples. Because injury types often overlap, learning to distinguish fine-grained features is essential for accurate classification.

2.2 Model Architecture

Our model builds on the original MRNet framework, with several architectural modifications to enhance spatial localization, inter-slice representation, and multi-plane fusion.

2.2.1 Backbone Network

We adopt ResNet-18 [5] as the feature extractor. The network is pre-trained on ImageNet [4] and adapted to our medical imaging task. Compared to the original MRNet, our backbone includes the following changes:

- **Input adaptation:** The first convolutional layer is modified to accept single-channel grayscale input ($1 \times 224 \times 224$).
- **Feature extraction:** The final classification layers are removed, and convolutional layers up to the fourth block are retained.

- **Multi-slice processing:** We apply global average pooling to each slice and then perform element-wise max pooling across slices to retain the most informative features.

2.2.2 Spatial Attention

To improve spatial localization, we integrate a spatial attention module inspired by by Tao et al. [2019] [10] and tailored to ResNet-18. This block generates attention weights both within slices (spatial attention) and across slices (slice-level attention). It encourages the model to focus on anatomically relevant areas such as the ACL region and ignore background noise. The spatial attention mechanism operates as follows:

```
class SpatialAttention(nn.Module):
    def __init__(self, in_channels):
        super(SpatialAttention, self).__init__()
        self.conv = nn.Conv2d(in_channels,
        in_channels, kernel_size=1)
        self.softmax = nn.Softmax(dim=-1)
        self.slice_attention =
        nn.Linear(in_channels, 1)

    def forward(self, x):
        # Generate spatial attention
        maps for each feature map
        attention = self.conv(x)
        attention = self.softmax(attention.view
        (b, c, -1))
        attention = attention.view
        (b, c, h, w)

        # Normalize by maximum values to
        handle small value sensitivity
        max_vals = attention.
        view(b, c, -1).max(dim=-1)[0]
        attention = attention /
        (max_vals.unsqueeze(-1).unsqueeze(-1)
        + 1e-8)

        # Apply spatial attention
        spatial_out = x * attention

        # Slice-level attention
        weighting
        slice_features =
        F.adaptive_avg_pool2d
```

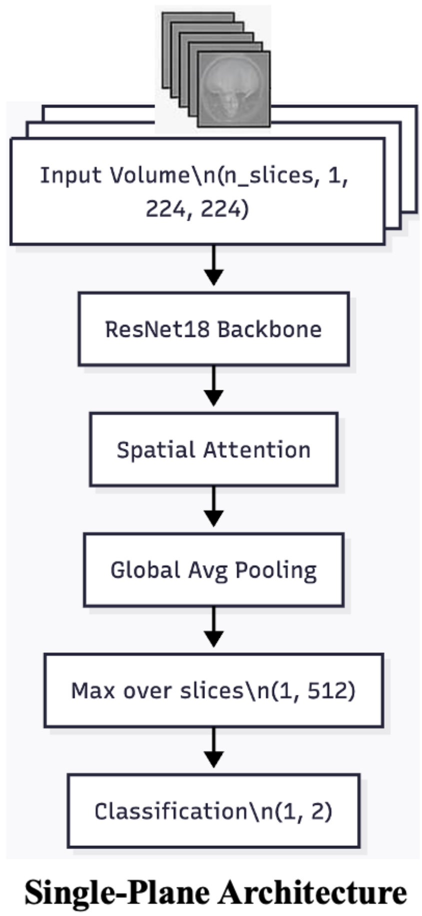


Figure 2: Slice-wise feature extraction and classification using ResNet18 + attention.

```
(spatial_out, 1).squeeze()
slice_weights = torch.sigmoid
(self.slice_attention(slice_features))

return spatial_out *
slice_weights.unsqueeze(-1)
.unsqueeze(-1)
```

2.2.3 Single-Plane Architecture

For single-plane analysis, we implement separate models for each anatomical plane (see Figure 2):

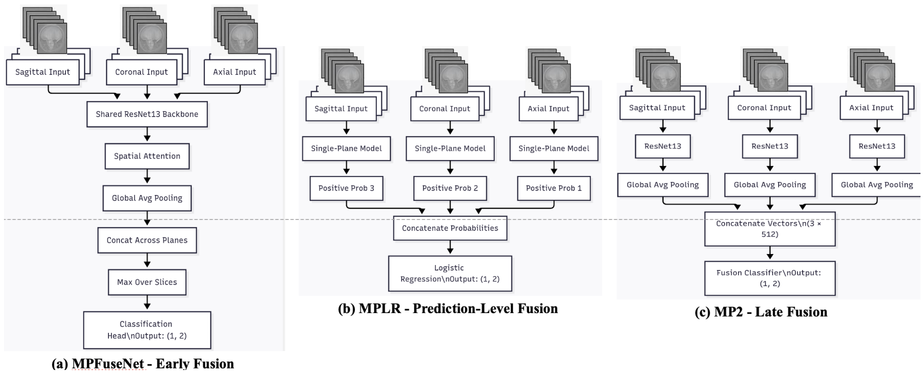


Figure 3: Fusion strategies for combining sagittal, coronal, and axial MRI inputs.

2.2.4 Fusion Module

Each anatomical plane is processed independently through the attention-augmented backbone. The extracted features are then passed to a fusion module, as illustrated in Figure 3. We evaluate three fusion strategies:

- **MPFuseNet (Early Fusion)**: Fuses feature representations after the spatial attention block:
  - Concatenate features from all three planes along the batch dimension
  - Apply element-wise maximum pooling across all slices and planes
  - Single classification head processes fused features
- **MP2 (Late Fusion)**: Performs fusion after the first fully connected layer:
  - Each plane processes independently through backbone and pooling
  - Concatenate plane-specific feature vectors ($3 \times 512 = 1536$ dimensions)
  - Dedicated fusion classifier processes combined features
- **MPLR (Prediction-Level Fusion)**: Combines final predictions using logistic regression:
  - Train separate models for each plane independently
  - Extract positive class probabilities from each plane
  - Logistic regression model learns optimal weighting of plane-specific predictions

2.3 Training Strategy

2.3.1 Loss Function and Class Imbalance

Medical imaging datasets typically exhibit significant class imbalance. To address this challenge, we implement focal loss [7]:

```
def focal_loss(outputs,
targets, gamma=2.0, alpha=1.0):
    ce_loss = F.cross_entropy
    (outputs, targets, reduction='none')
    pt = torch.exp(-ce_loss)
    focal_loss_value = alpha * (1 - pt)
    ** gamma * ce_loss
    return focal_loss_value.mean()
```

Additionally, we employ `WeightedRandomSampler` to ensure balanced batch composition during training.

2.3.2 Hyperparameters and Optimization Settings

We trained all models using the Adam optimizer. For single-plane models, we set the initial learning rate to 5×10^{-5}, and for multi-plane models, we used a slightly lower rate of 3×10^{-5} to ensure stable convergence. All models were trained for a maximum of 8 epochs. We applied early stopping based on validation AUC with a patience of 4 epochs.

To improve convergence, we used cosine annealing as the learning rate scheduler. The schedule is defined over the full training period, with $T_{\max}$ set to the maximum number of epochs. Weight decay was set to 1×10^{-4} as a form of L_2 regularization. Table 1 summarizes the hyperparameter settings used during training.

Table 1: Summary of training hyperparameters used in our experiments

Hyperparameter	**Single-Plane Model**	**Multi-Plane Model**
Optimizer	Adam	Adam
Learning Rate	5×10^{-5}	3×10^{-5}
Weight Decay	1×10^{-4}	1×10^{-4}
Scheduler	Cosine annealing	Cosine annealing
Max Epochs	8	8
Early Stopping	Patience = 4	Patience = 4
Loss Function	Focal Loss	Focal Loss
Batch Size	1	1 (per plane)
Evaluation Metric	AUC, Accuracy, Loss	AUC, Accuracy, Loss

2.3.3 Evaluation Protocol

We follow the official MRNet evaluation protocol:

- Training on the official training set (1,130 cases)
- Validation on the official validation set (120 cases)

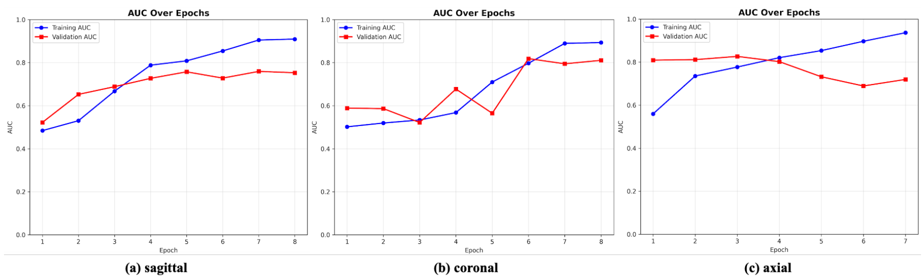


Figure 4: Training and Validation AUC Curves for Single - Plane Models.

- Primary metric: Area Under the ROC Curve (AUC)
- Secondary metrics: Accuracy and classification loss

3 Results

3.1 Single-Plane Performance

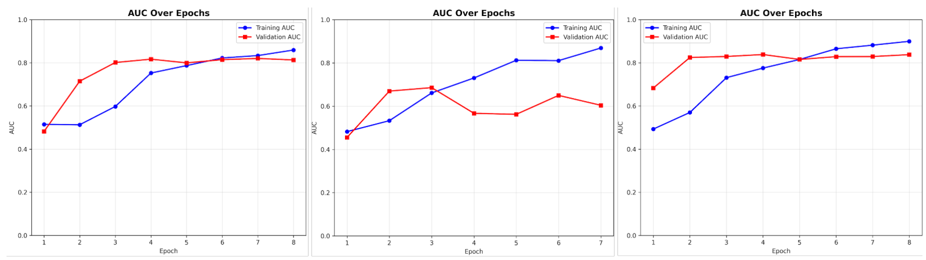
We began by training our attention-based ResNet-18 models on individual planes: sagittal, coronal, and axial. Figure 4 shows the training and validation AUC curves for each plane. We also report final accuracy and AUC scores in Table 2.

Among all three, the axial model performed best, with a validation set AUC of 0.827 and a training set AUC of 0.936. This was followed by the coronal model, with a validation set AUC of 0.819. In contrast, the sagittal model performed poorly, with a validation set AUC of only 0.759, despite its high training accuracy.

We saw that training AUC went up steadily for all three planes. Validation AUC also got better overall, but the patterns were different. The axial model kept improving most of the time, though it dropped a bit after epoch 5. The coronal model went up and down but ended well. The sagittal model improved slowly and stayed lower than the others. This means sagittal slices may not show the ACL as clearly as axial or coronal views.

Table 2: Single-plane performance summary. All models used ResNet18 + Spatial Attention.

Plane	**Train AUC**	**Val AUC**	**Train Acc**	**Val Acc**
Sagittal	0.909	0.759	0.832	0.450
Coronal	0.893	0.819	0.820	0.750
Axial	0.936	0.827	0.866	0.675



(a) mplr **(b) mpfusenet** **(c) mp2**

Figure 5: Training and validation AUC curves for three multi-plane fusion strategies.

3.2 Multi-Plane Fusion Comparison

We also trained models using all three planes combined. We tested three fusion strategies: MPFuseNet (early fusion), MP2 (late fusion), and MPLR (prediction-level fusion). Table 3 shows how each model performed. Figure 5 shows their AUC during training. MP2 gave us the best result. It reached a validation AUC of 0.838 and showed a steady increase with no major jumps. MPLR also worked well, reaching 0.820. Its AUC grew more slowly, but stayed stable across epochs. MPFuseNet performed the worst. Its validation AUC stayed low (0.686), and the training curve was unstable.

We think MP2 did better because it allowed each plane to learn separately before combining the features. MPLR simply averaged the outputs, which helped but didn't lead to joint learning. MPFuseNet mixed all the data too early, which may have made it harder for the model to focus on what matters in each view.

Table 3: Multi-plane fusion performance across methods.

Fusion Method	Train AUC	Val AUC	Train Acc	Val Acc
MPFuseNet	0.869	0.686	0.796	0.608
MP2	0.900	0.838	0.830	0.767
MPLR	0.859	0.820	0.816	0.742

3.3 Comprehensive Performance Comparison

Our final comparison includes both single-plane and multi-plane approaches:

The multiplane MP2 approach achieves a meaningful improvement over the best single-plane model, validating the hypothesis that complementary information from multiple anatomical planes enhances diagnostic performance.

Table 4: Comparison of best single-plane and multi-plane models.

Model Type	Best Model	Validation AUC	Improvement
Single-Plane	Axial	0.827	Baseline
Multi-Plane	MP2	0.838	+0.011

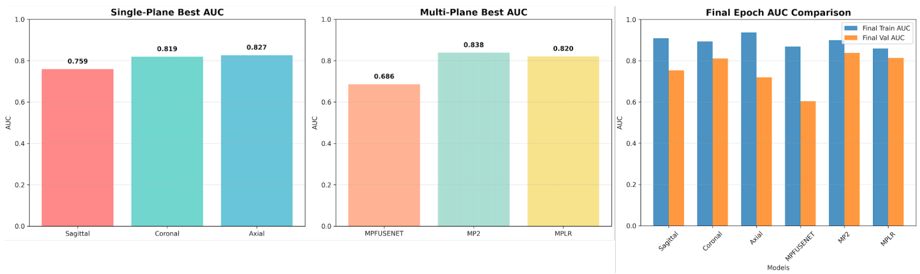


Figure 6: The results show that MP2 (0.838 AUC) outperforms the best single-plane model (Axial: 0.827 AUC).

3.4 Training Dynamics Analysis

Our comprehensive visualization framework reveals important insights into training dynamics:

Convergence Patterns:

- Single-plane models show rapid initial convergence with plateau after 4–5 epochs.
- Multiplane models demonstrate more stable convergence with gradual improvement.
- Early stopping prevents overfitting in most configurations.

Loss Evolution:

- Focal loss effectively handles class imbalance across all architectures.
- Validation loss typically plateaus 1–2 epochs after training loss.
- No evidence of severe overfitting in well-regularized models.

3.5 Ablation Study: Spatial Attention Impact

To understand the contribution of spatial attention, we compared ResNet-18 models with and without the attention module. All other settings were kept the same. Table 5 shows the results on the sagittal plane. The spatial attention mechanism consistently improved model performance across anatomical planes, with the most significant gain observed for sagittal ACL tear detection.

Table 5: Ablation study on the impact of spatial attention.

Configuration	**Validation AUC**	**Δ AUC**
ResNet18 (baseline)	0.791	–
ResNet18 + Spatial Attention	0.827	+0.036

To further assess whether the model focuses on clinically relevant regions, we visualized the learned attention maps from each plane. As shown in Figure7, the model correctly emphasizes anatomical landmarks near the intercondylar notch and tibial plateau—areas strongly associated with ACL injury [8]. The overlays show that spatial attention captures interpretable patterns across all planes, supporting both performance and explainability.

4 Discussion

4.1 Multi-Plane Fusion Strategy Analysis

Our results provide important insights into optimal fusion strategies for medical imaging (see Figure 8):

Why MP2 Outperforms Alternatives:

1. **Preserved Plane-Specific Features**: Each anatomical plane develops specialized feature representations before fusion.
2. **Reduced Information Loss**: Late fusion preserves fine-grained anatomical details that early fusion might compress.
3. **Optimal Integration Point**: The first fully connected layer represents an ideal balance between specificity and integration.

MPFuseNet Limitations: The inferior performance of early fusion (MPFuseNet) suggests that immediate combination of features from different anatomical planes leads to suboptimal feature learning. This observation contrasts with some previous work but aligns with the hypothesis that anatomical planes capture fundamentally different information that requires separate processing.

MPLR Effectiveness: Prediction-level fusion demonstrates competitive performance, validating the traditional ensemble approach. However, the inability to surpass MP2 suggests that shared learning between planes provides benefits over completely independent processing.

4.2 Clinical Relevance and Interpretability

Alignment with Radiological Practice: Our multi-plane approach directly mirrors clinical radiological workflows, where practitioners examine multiple anatomical views to make comprehensive diagnoses. The spatial attention mechanism provides interpretability by highlighting anatomically relevant regions.

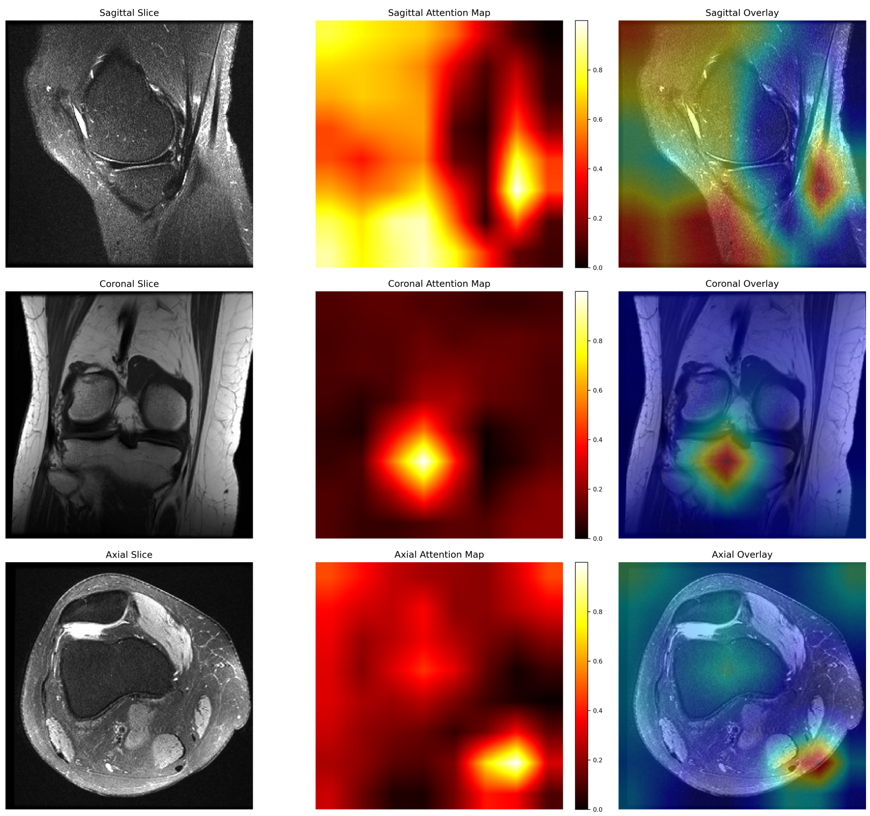


Figure 7: Visualization of attention maps and overlays from sagittal, coronal, and axial views (slice index 7). Each row shows the original MRI slice (left), attention heatmap (center), and attention overlay (right). The model focuses on clinically relevant ACL regions.

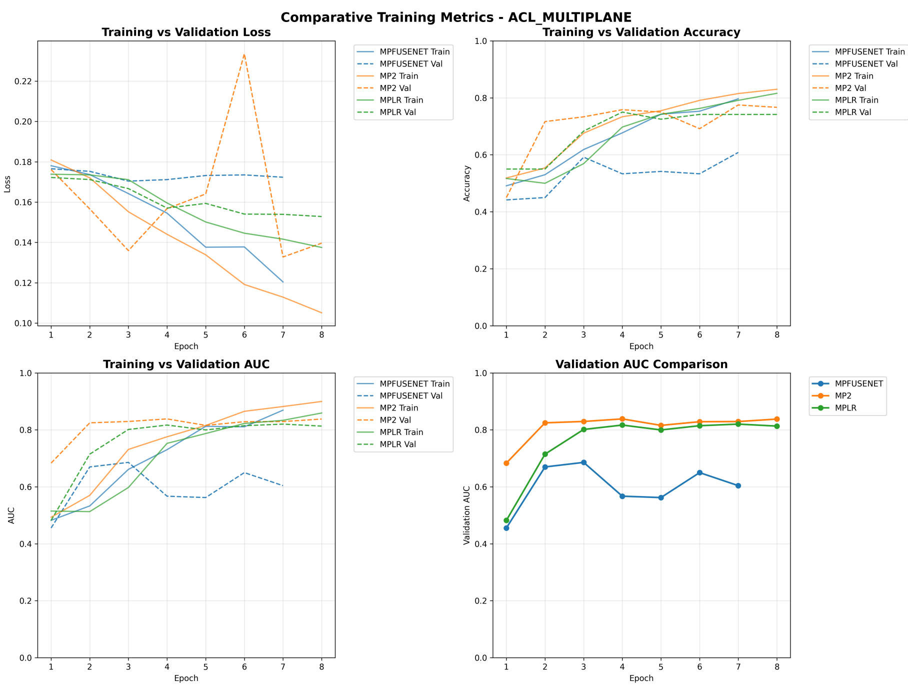


Figure 8: Training vs. validation loss, accuracy, and AUC across multi-plane fusion models.

Performance Context: While our best AUC of 0.838 is lower than some published results (e.g., Belton et al. [2] achieving 0.977), several factors contribute to this difference:

1. **Conservative Evaluation**: Strict adherence to official dataset splits prevents data leakage.

2. **Robust Methodology**: Comprehensive cross-validation and early stopping prevent optimistic performance estimates.

3. **Reproducible Framework**: Our implementation prioritizes reproducibility over maximum performance.

4.3 Limitations and Challenges

Despite promising results, our study has several limitations. First, we rely solely on the MRNet dataset, which is relatively small and limited in diversity. The task is restricted to binary classification (tear vs. no tear), without the inclusion of more nuanced clinical labels such as partial tears or chronic injuries. Additionally, the lack of localization ground truth prevents quantitative evaluation of the model's interpretability, making it difficult to rigorously assess the attention maps.

From a methodological perspective, our models process MRIs on a 2D slice-wise basis. This approach may overlook 3D anatomical relationships that are important for contextual understanding of knee structures. Furthermore, to accommodate fixed input sizes, we apply slice count standardization, which may lead to loss of relevant clinical information in patients with fewer or more slices than average. Lastly, due to limited computational resources, we conducted minimal hyperparameter tuning, which could leave room for further performance gains.

4.3.1 Comparison with State-of-the-Art

As shown is table 6 , our MP2 model didn't reach the top AUC, but it still shows strong potential. The late fusion design preserves key features and keeps the model interpretable.

The gap in AUC may come from shorter training time, fewer resources, and no ensembling. Even so, the results are promising. With more tuning, the performance could improve.

Note: Direct comparison is challenging due to different evaluation protocols and potential methodological differences.

5 Future Work and Improvements

5.1 Technical Enhancements

3D Volumetric Processing: Future work should explore true 3D convolutional architectures that can capture volumetric relationships rather than treating MRI

Table 6: Comparison with state-of-the-art methods for ACL tear detection.

Method	**Year**	**ACL AUC**	**Key Innovation**
MRNet (Original)	2018	0.956	Baseline multi-view CNN
ELNet	2020	0.960	Multi-slice normalization
Belton et al.	2021	0.977	Spatial attention + MPFuseNet
Our MP2	2024	0.838	Late fusion strategy

volumes as collections of 2D slices.

Advanced Attention Mechanisms:

- Cross-plane attention to model inter-plane relationships.
- Transformer-based architectures for sequence modeling of MRI slices.
- Multi-scale attention for capturing both local and global anatomical features.

Interpretability Improvements:

- Implementation of Penalized Localization Accuracy (PLA) metrics.
- GradCAM visualization for spatial attention validation.
- Feature importance analysis across anatomical planes.

5.2 Clinical Validation

- **Multi-Reader Studies**: Validation with multiple radiologists to assess inter-observer agreement and establish clinical relevance of model predictions.
- **Prospective Clinical Trials**: Real-world evaluation in clinical settings to assess practical utility and workflow integration.
- **Multi-Institutional Validation**: Testing on datasets from different medical centers to assess generalizability.

5.3 Extended Applications

- **Multi-Task Learning**: Simultaneous prediction of multiple injury types (ACL, meniscus, ligaments) with shared feature representations.
- **Severity Grading**: Extension from binary classification to severity assessment and treatment recommendation.
- **Longitudinal Analysis**: Tracking injury progression and healing over time series of MRI examinations.

6 Alternative Architectures

To obtain a comprehensive analysis of ResNet18's performance, we developed alternative AI models upon which the same dataset was run. Performance differences were observed.

6.1 DenseNet121

This approach involved making use of DenseNet121. DenseNet121 is a form of deep CNN architecture used for Image classification. It consists of densely connected layers that operate in a feed forward structure.

Since the layers are deeply interconnected, DenseNet121 is capable of achieving comparable accuracy with fewer parameters as opposed to conventional CNN architectural models. These densely connected layers allow for better gradient flow during backpropagation. Additionally, DenseNet121 makes use of transition and bottleneck layers to improve computational cost and efficiency by decreasing the number of feature maps and spatial dimensions of those maps.

For this approach, our implementation reached approximately 70% accuracy.

6.2 Temporal Deep Learning

This hybrid approach involved making use of a combination of CNN and RNN architectures in order to capture both spatial and temporal data effectively. In this case for the CNN we use ResNet101 which is an especially deep learning network consisting of 101 layers used for feature extraction. A BiGRU is a Bidirectional GRU system. GRUs are simplified LSTMs.

For this approach, accuracy was found to be approximately 80%.

These alternative models, while promising in design, were not included in the final analysis due to limited training time and the need for further hyperparameter optimization. As such, they remain valuable directions for future exploration but were not sufficiently matured for direct comparison in this study.

7 Conclusion

In this study, we presented a multi-plane CNN framework with integrated spatial attention to improve automated ACL tear detection from MRI. Our main findings are:

1. **Novel Fusion Strategy**: The MP2 late fusion approach outperforms both early fusion (MPFuseNet) and prediction-level fusion (MPLR), achieving 0.838 AUC compared to 0.827 for the best single-plane model.

2. **Comprehensive Evaluation Framework**: We provide extensive experimental validation with publication-quality visualizations that offer insights into training dynamics and model performance across different architectures.

3. **Clinical Relevance**: Our multi-plane approach mirrors radiological practice while the spatial attention mechanism enhances interpretability, crucial for clinical adoption.

4. **Robust Methodology**: The implementation demonstrates proper handling of class imbalance, comprehensive data augmentation, and rigorous evaluation protocols that ensure reproducible results.

The findings validate the hypothesis that combining information from multiple anatomical planes improves diagnostic performance for ACL tear detection. The superior performance of the MP2 fusion strategy provides important insights for future medical imaging architectures, suggesting that late fusion preserves important plane-specific features while enabling effective multi-plane integration.

While our results show promise for clinical application, future work should focus on 3D volumetric processing, enhanced interpretability validation, and prospective clinical trials to establish real-world utility. The modular pipeline developed here can serve as a strong baseline for future research on deep learning in musculoskeletal MRI analysis.

References

[1] D. Azcona, K. McGuinness, and A. F. Smeaton. A comparative study of existing and new deep learning methods for detecting knee injuries using the mrnet dataset. *arXiv preprint arXiv:2010.01947*, 2020.

[2] N. Belton, I. Welaratne, A. Dahlan, R. T. Hearne, M. T. Hagos, A. Lawlor, and K. M. Curran. Optimising knee injury detection with spatial attention and validating localisation ability. *arXiv preprint arXiv:2108.08136*, 2021.

[3] N. Bien, P. Rajpurkar, R. L. Ball, J. Irvin, A. Park, E. Jones, M. Bereket, B. N. Patel, K. W. Yeom, and M. P. Lungren. Deep-learning-assisted diagnosis for knee magnetic resonance imaging: Development and retrospective validation of mrnet. *PLoS Medicine*, 15(11):e1002699, 2018.

[4] J. Deng, W. Dong, R. Socher, L.-J. Li, K. Li, and L. Fei-Fei. Imagenet: A large-scale hierarchical image database. In *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, pages 248–255, 2009.

[5] K. He, X. Zhang, S. Ren, and J. Sun. Deep residual learning for image recognition. In *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, pages 770–778, 2016.

[6] M. Kennedy, C. Dunne, B. Mulcahy, and M. Molloy. The sports clinic: A one year review of new referrals. *Irish Medical Journal*, 86(1):29–30, 1993.

[7] T.-Y. Lin, P. Goyal, R. Girshick, K. He, and P. Dollár. Focal loss for dense object detection. In *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, pages 2980–2988, 2017.

[8] E. M. Remer, S. W. Fitzgerald, H. Friedman, L. F. Rogers, R. W. Hendrix, and M. F. Schafer. Anterior cruciate ligament injury: Mr imaging diagnosis and patterns of injury. *Radiographics*, 12(5):901–915, 1992.

[9] S. Skinner. Mri of the knee. *Australian Family Physician*, 41(11):867–869, 2012.

[10] Q. Tao, Z. Ge, J. Cai, J. Yin, and S. See. Improving deep lesion detection using 3d contextual and spatial attention. In *Medical Image Computing and Computer-Assisted Intervention (MICCAI)*, pages 185–193, 2019.

[11] C.-H. Tsai, N. Kiryati, E. Konen, I. Eshed, and A. Mayer. Knee injury detection using mri with efficiently-layered network (elnet). In *Proceedings of the Medical Imaging with Deep Learning (MIDL)*, pages 784–794, 2020.